Laurent Juliá Calac



Data Scientist | AI & Machine Learning | M.S. Data Science & Analytics

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Portfolio | LinkedIn | GitHub
Location: Washington, D.C., USA (studies) | Reutlingen, Germany (home)
Nationality: German (EU citizen, no work permit required in the EU)

PROFILE

Data science and AI master's student at Georgetown University (GPA 4.0/4.0) with hands-on experience building ETL pipelines, BI dashboards, machine learning models, and LLM prototypes in Germany, the Dominican Republic, and the USA. Skilled in Python, SQL, deep learning, NLP, and generative AI, with a focus on translating business needs into practical, scalable data and AI solutions and communicating results to technical and non-technical stakeholders. Multilingual; former NCAA Division I student-athlete. Available for full-time positions in Europe from June 2027.

WORK EXPERIENCE

Foerster Group, Reutlingen, Germany 06/2026 – 08/2026
AI & Data Science Intern

- Architected an end-to-end **ETL pipeline** in Python, SQL Server, and SQLite that unified 20 years of inspection data (**4,000+ files from 2,000+ inspections**) from database backups, Excel defect files, and PDF/Word reports into one data platform.
- Automated SQL Server backup ingestion with a dynamic schema that auto-discovers tables and adapts to schema drift across software versions, eliminating manual database restoration.
- Built an Excel parsing engine handling multi-row headers and shifting layouts across 6 inspection types, converting inconsistent spreadsheets into structured, analysis-ready records.
- Developed a **Streamlit dashboard** prototype with drill-down filters and automated KPI/status logic, giving a team of 10 inspection engineers a single view of every inspection; gathered requirements with the engineering department and demoed the prototype ahead of rollout.
- Benchmarked general-purpose and text-to-SQL **LLMs** (Ollama, Hugging Face) on execution accuracy, SQL validity, and latency for natural-language querying of the database; top models reached **70% execution accuracy**, and recommendations presented to the team and manager guided the planned LLM integration.

Grupo Mallén, Santo Domingo, Dominican Republic 07/2025 – 08/2025
Data Analytics Intern

- Led BI consolidation for the CFO and senior management, reducing **20 Power BI dashboards to 5** executive-ready views and cutting reporting complexity by **75%** while maintaining data integrity.
- Partnered with the CFO and marketing team to define requirements for sales, marketing, and budgeting dashboards that accelerated decision-making across departments.

Georgia Southern University, Statesboro, USA 05/2025 – 07/2025
Data Analytics Intern

- Automated weekly IT help-ticket reporting with Python, SQL, Excel, and Power BI, tracking ticket volume, resolver, and resolution time.
- Built Power BI dashboards tracking invoice volume, cycle time, on-time rate, exceptions, and staffing for system leadership.

EDUCATION

Georgetown University, Washington, D.C., USA 08/2025 – 05/2027 (expected)
M.S. in Data Science & Analytics, Concentration in Artificial Intelligence | GPA 4.0/4.0

- Relevant coursework: Neural Networks & Deep Learning, Natural Language Processing, Reinforcement Learning, Applied Generative AI, Big Data & Cloud Computing, Statistical Learning.

Georgia Southern University, Statesboro, USA 01/2022 – 07/2025
B.S. in Information Technology | GPA 3.75/4.0, Magna Cum Laude

- Focus areas: programming, databases, data analysis, information and network security.

Albert-Einstein-Gymnasium, Reutlingen, Germany 09/2013 – 07/2021
Abitur (German university entrance qualification), bilingual English stream

- Majors: Mathematics, English, Sports.

PROJECTS

ECG Arrhythmia Classification | Deep Learning 04/2026

- Developed and compared CNN, Transformer, and hybrid models for 4-class ECG arrhythmia classification, achieving **99.0% test accuracy and 0.93 macro F1**; Transformers on raw waveforms raised macro F1 from 0.71 to 0.86 over CNNs.
- Designed preprocessing pipelines for waveforms, spectrograms, and metadata that addressed class imbalance.

Gender Bias in Wikipedia Biographies | NLP & Fairness Research 12/2025

- Analyzed **573K+ biographies** with a modular, tested NLP pipeline (scraping, preprocessing, inference, analysis; Poetry, pytest, multiprocessing) using VADER, TextBlob, and RoBERTa; identified statistically significant gender differences via t-, KS-, Mann–Whitney U, and χ^2 tests.

AI-Powered Tennis Analytics Tool | Computer Vision & LLMs, Bachelor's Capstone 2025

- Built a YOLO-based computer vision system tracking shot speed, player movement, and positioning from match video, connected via an API pipeline to a locally hosted LLM and a dashboard for video upload, automated analysis, and interactive match insights.

SKILLS

Programming & Data: Python (Pandas, Polars, NumPy, scikit-learn), SQL, R

Machine Learning & AI: PyTorch, TensorFlow, Hugging Face, LLMs, LLM Evaluation, Agentic AI, NLP, Computer Vision (YOLO), CNNs, Transformers

Data Engineering & BI: ETL, SQL Server, PostgreSQL, MySQL, SQLite, DuckDB, Parquet, Spark, AWS (EC2), Streamlit, Power BI, Git/GitHub, CI/CD

Languages: German (native), Spanish (native), English (fluent, full professional proficiency), French (limited working proficiency)

LEADERSHIP & ACTIVITIES

Division I Student-Athlete, Georgia Southern Men's Tennis Team Captain 01/2022 – 05/2025

- Balanced a 20-hour weekly training and national competition schedule with full-time studies (Magna Cum Laude); helped lead the team to its first-ever Sun Belt Regular Season Conference championship and the program's highest national ranking (No. 63).

Student-Athlete Advisory Committee Representative, Georgia Southern University 08/2023 – 05/2025

- Represented the men's tennis team and organized fundraising initiatives that raised $1,500+ for cancer awareness campaigns.

Graduate Student Mentor, Georgetown M.S. Data Science & Analytics 09/2026 – present

- Mentor a first-year student on academics, technical development, and career opportunities.

Volunteer Graduate Assistant, Georgetown Men's & Women's Tennis 09/2025 – present

- Provide tactical coaching and match analysis during practices and competitions; support donor engagement and fundraising.